

Java Programming Notes

A concise reference document for testing a Retrieval-Augmented Generation (RAG) chatbot.

1. What is Java?

Java is a high-level, general-purpose programming language designed to be portable and object-oriented. Java source code is compiled into bytecode, which can run on a Java Virtual Machine (JVM).

2. Features of Java

Important features include platform independence, object-oriented programming, automatic memory management, strong type checking, exception handling, multithreading, security features, and a large standard library.

3. JVM, JRE and JDK

JVM (Java Virtual Machine) executes Java bytecode. JRE (Java Runtime Environment) provides the JVM and runtime libraries needed to run Java applications. JDK (Java Development Kit) contains development tools such as the Java compiler along with the runtime components.

4. Variables and Data Types

A variable stores a value. Java has primitive data types such as byte, short, int, long, float, double, char, and boolean. Java also supports reference types such as classes, arrays, and interfaces.

5. Operators

Java provides arithmetic operators such as +, -, *, / and %. Relational operators include ==, !=, >, <, >= and <=. Logical operators include &&, || and !. Assignment operators include =, +=, -=, *= and /=.

6. Conditional Statements

The if statement executes code when a condition is true. The if-else statement selects between two paths. The else-if ladder handles multiple conditions. The switch statement selects a block based on an expression.

7. Loops

The for loop is commonly used when the number of iterations is known. The while loop checks its condition before each iteration. The do-while loop executes its body at least once. break exits a loop and continue skips to the next iteration.

8. Arrays

An array stores multiple values of the same type in a fixed-size structure. Array indexes start at 0. For an array arr, arr.length gives its number of elements. Example: `int[] numbers = {10, 20, 30};`

9. Strings

String is a class used to represent text. String objects are immutable, meaning their contents cannot be changed after creation. Common methods include `length()`, `charAt()`, `substring()`, `equals()`, and `toUpperCase()`.

10. Object-Oriented Programming

The main object-oriented concepts are encapsulation, inheritance, polymorphism, and abstraction. A class defines the structure and behavior of objects, while an object is an instance of a class.

11. Inheritance

Inheritance allows one class to acquire accessible fields and methods from another class. In Java, a class can extend another class using the `extends` keyword. Inheritance supports code reuse and hierarchical design.

12. Polymorphism

Polymorphism means that one interface or method call can represent different behaviors. Method overloading is compile-time polymorphism, while method overriding is commonly associated with runtime polymorphism.

13. Encapsulation

Encapsulation means keeping data and the methods that operate on that data together and controlling access to internal state. Private fields combined with public getter and setter methods are a common implementation.

14. Abstraction

Abstraction focuses on essential behavior while hiding implementation details. Java supports abstraction through abstract classes and interfaces.

15. Exception Handling

Exceptions represent abnormal conditions during program execution. Java provides `try`, `catch`, `finally`, `throw`, and `throws` for exception handling. Checked exceptions are checked by the compiler, while unchecked exceptions include runtime exceptions.

16. Collections

The Java Collections Framework provides reusable data structures and algorithms. Common interfaces and classes include `List`, `Set`, `Map`, `ArrayList`, `LinkedList`, `HashSet`, `TreeSet`, `HashMap`, and `TreeMap`.

17. Multithreading

Multithreading allows multiple threads to execute concurrently within a program. Threads can be created using the `Thread` class or by implementing `Runnable`. Synchronization can be used to control access to shared data.

18. Sample Java Program

Example: `public class Main { public static void main(String[] args) { System.out.println("Hello, Java!"); } }`

The main method is the entry point used to start a standard Java application.